Andrew Lott
University of Georgia
Department of Mathematics, Boyd 434J, Athens, Georgia

September 16, 2026

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Education

- **University of Georgia** — Georgia, USA
  Ph.D. candidate in Mathematics — *2023–present*
  - Expected graduation: May 2027.
  - Advisors: Ákos Magyar and Giorgis Petridis.
- **Millsaps College** — Mississippi, USA
  B.S. in Mathematics, summa cum laude — *2023*

Appointments

- **Graduate Teaching Assistant**
  University of Georgia, Department of Mathematics. Instructor of record and grader.
  August 2023–present.
- **Visiting Researcher**
  Alfréd Rényi Institute of Mathematics, Budapest, Hungary.
  Simons Semester in New Directions in Modern Harmonic Analysis and Applications.
  January–June 2026.

Research Interests

- Additive combinatorics, analytic number theory, and ergodic theory; the circle method, exponential sum estimates, forbidden difference problems, higher-order Fourier analysis, transference, pointwise ergodic theorems, Ramsey theory, sieve methods, and function fields.

Publications and Preprints

- Alexandros Kalogirou, Andrew Lott, Ákos Magyar, and Akash Singha Roy. *The Furstenberg–Sárközy theorem for sums of an even number of odd powers.* Preprint, 2026. arXiv:2609.16595.
- Andrew Lott, Ákos Magyar, and Nagendar Reddy Ponagandla. *Quasipolynomial density bounds for K-point configurations in $\mathbb{Z}^d$.* Submitted, 2026. arXiv:2609.13126.
- Swaroop Hegde, Andrew Lott, Giorgis Petridis, and Nagendar Reddy Ponagandla. *Refined upper bounds on Schur-like numbers.* Submitted, 2026. arXiv:2608.03661.
- Steve Fan and Andrew Lott. *The van der Corput property for sums of two squares.* Submitted, 2026. arXiv:2606.29185.
- Thái Hoàng Lê and Andrew Lott. *Bourgain's L^2 pointwise ergodic theorem over function fields.* Submitted, 2026. arXiv:2605.28997.
- Carlo Francisco E. Adajar, Rishika Agrawal, Mukul Rai Choudhuri, Chian Yeong Chuah, Steve Fan, Swaroop Hegde, Andrew Lott, Krishnamohan Nandakumar, and Nagendar Reddy Ponagandla. *Extensions of the Furstenberg–Sárközy theorem via the arithmetic level-d inequality.* Submitted, 2026. arXiv:2605.16216.

- Steve Fan and Andrew Lott. *Sárközy's theorem in $\mathbb{F}_q[t]$ via the van der Corput property. Finite Fields and Their Applications* **116** (2026), Article 102899. arXiv:2510.27581.
- Andrew Lott, Ákos Magyar, Giorgis Petridis, and János Pintz. *Polynomial configurations in dense subsets of the prime lattice. Acta Arithmetica*, to appear. arXiv:2504.14424.
- Andrew Lott and Nagendar Reddy Ponagandla. *Polynomial progressions in the generalized twin primes. Proceedings of the American Mathematical Society* (2026). arXiv:2505.17375.

Talks

- *Two Sárközy-type results.* Recent Trends in Ergodic Theory, Symbolic Dynamics, and Connections to Combinatorics, Northwestern University, 2026.
- *K-point configurations in the integer lattice.* Discrete Analysis Seminar, HUN-REN Alfréd Rényi Institute of Mathematics, 2026.
- *Extensions of the Furstenberg–Sárközy theorem via the arithmetic level-d inequality.* Number Theory Seminar, HUN-REN Alfréd Rényi Institute of Mathematics, 2026.
- *Sárközy's theorem in function fields.* Number Theory Seminar, HUN-REN Alfréd Rényi Institute of Mathematics, 2026.
- *Sárközy's theorem in function fields.* Number Theory Seminar, University of Georgia, 2025.
- *Sárközy's theorem in function fields.* Number Theory Seminar, University of Mississippi, 2025.
- *Sárközy's theorem in function fields.* AMS Fall Southeastern Sectional Meeting, Tulane University, 2025.
- *Polynomial configurations in the primes.* INTEGERS 2025 Conference, University of Georgia, 2025.
- *Polynomial configurations in the primes.* Kinnaird Summer Research Program, Millsaps College, 2025.
- *The sum-product problem for small sets.* Graduate Student Seminar, University of Georgia, 2024.
- *Schur's theorem in integer lattices.* AMS Spring Southeastern Sectional Meeting, Georgia Tech, 2023.
- *Schur's theorem in integer lattices.* INTEGERS 2023 Conference, University of Georgia, 2023.
- *Notes and computations on congruence covering systems.* MAA Louisiana–Mississippi Sectional Meeting, University of Mississippi, 2023. First place student presentation.
- *Notes and computations on congruence covering systems.* TriBeta Symposium, Millsaps College, 2023. First place student presentation.
- *New and old results in Ramsey theory and additive combinatorics.* Honors Research Conference, Millsaps College, 2023. Award for best honors thesis.
- *Schur's theorem in integer lattices.* MAA Louisiana–Mississippi Sectional Meeting, Northwestern State University of Louisiana, 2022. First place student presentation.

Teaching

- **MATH 1113 — Precalculus** — University of Georgia
  Instructor of record — *Fall 2024, 2025, 2026*
- **MATH 2250 — Calculus I** — University of Georgia
  Instructor of record — *Spring 2025*
- **MATH 8100 — Measure Theory** — University of Georgia
  Grader; graduate qualifying exam course — *Fall 2025*
- **MATH 3500H — Honors Multivariable Calculus** — University of Georgia
  Grader — *Fall 2024*

- **MATH 4690/6690 — Graph Theory** University of Georgia
  Grader *Spring 2024*
- **MATH 3100 — Introduction to Real Analysis** University of Georgia
  Grader *Fall 2023*
- **Calculus II** Millsaps College
  Teaching assistant *Spring 2023*
- **Calculus I** Millsaps College
  Teaching assistant *Fall 2022*

Advising and Mentorship

- **Directed Reading Program** University of Georgia
  Coordinator *Spring and Fall 2026*
- **Directed Reading Program** University of Georgia
  Undergraduate mentor *Spring and Fall 2025*
  - Mentored Priya Jani in prime number theory, Fall 2025, using Paul Pollack's *Not Always Buried Deep*.
  - Mentored José López-Gervacio on Roth's theorem, Spring 2025, using Yufei Zhao's *Graph Theory and Additive Combinatorics*.
- **Kinnaird Undergraduate Research Program** Millsaps College
  Teaching assistant *Summers 2023, 2024*

Service and Outreach

- **UGA High School Math Competition** University of Georgia
  Volunteer and problem writer *Fall 2025*
- **Athens Math Circle** *Spring 2025*
  Presented an introduction to Ramsey numbers for middle and high school students.
- **Jackson Preparatory School MATHCOUNTS Team**
  Assistant coach *2022*
- **Millsaps College High School Math Competition**
  Volunteer and problem writer *2021, 2022*